LEARNER GUIDE

For

Agentic AI for Social Media Marketing

TGS-2020505996

TSC: Social Media Marketing · HAS-OTO-4007-1.1

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v15.0 · 11 September 2026

Document Version Control Record

Version Number	Effective Date	Summary of Included Changes	Author
v15.0	11 September 2026	Rebuilt from the approved v14 Facebook deck; expanded with agentic architecture, governance, current controls, ten self-contained labs and mock data.	Dr Alfred Ang

Table of Contents

Document Version Control Record....3
Table of Contents....3
1. Course Overview....4
Learning Outcomes....4
Assessment....4
2. Responsible Agentic Marketing Operating Model....4
3. Topic 01: Plan an Agentic Social Media Campaign....4
Translate Business Goals into a KPI Tree....4
Encode a Persona as Evidence, Not Fiction....5
Model the Buyer Journey as Observable States....5
Build a Social Media Audit Inventory....6
Score Marketing Opportunities Transparently....6
Design the Marketing Agent Architecture....6
Write a Tool Contract the Agent Can Obey....7
Set Policy Boundaries Before Automation....7
Lab 1: Build an Agent-Ready Campaign Charter....7
Lab 2: Create an Evidence-Based Persona and Journey Agent....9
4. Topic 02: Create Campaigns with Governed Agents....10
Represent Campaign, Ad Set, and Ad....10
Turn a Creative Brief into Structured Inputs....11
Encode Brand Voice as Testable Rules....11
Build Audiences with Purpose Limitation....11
Generate Content Through a Review Pipeline....12
Place a Human Approval Gate Before Publish....12
Choose Placement and Format with a Matrix....12
Design a Clean Campaign Experiment....13
Lab 3: Build a Brand-Safe Content Brief Agent....13

Lab 4: Assemble a Governed Campaign Package....15
5. Topic 03: Measure Performance and Agent Decisions....16
Map the Measurement Event Pipeline....16
Calculate Reach, CTR, CVR, CPL and ROAS....17
Build a Funnel with Diagnostic Rates....17
Separate Attribution from Incrementality....17
Design a Reporting Schema for Agents....18
Detect Anomalies with Thresholds and Context....18
Build Retargeting with Consent and Suppression....19
Observe the Agent, Not Just the Campaign....19
Lab 5: Compute a Campaign Measurement Scorecard....19
Lab 6: Build a Listening and Anomaly Triage Agent....21
6. Topic 04: Optimize Campaigns with Controlled Experiments....22
Run the Observe–Diagnose–Act Loop....22
Use Practical Significance in A/B Decisions....23
Reallocate Budget with Marginal Return....23
Detect Creative Fatigue....23
Control Audience Overlap and Frequency....24
Schedule Content with Dependencies....24
Compare Platforms with a Feasibility Matrix....25
Define Rollback and Kill-Switch Rules....25
Lab 7: Evaluate an A/B Test and Budget Decision....25
Lab 8: Orchestrate an Approval-Aware Content Calendar....27
7. Topic 05: Manage Communities with Human Oversight....28
Route Community Messages by Intent and Risk....28
Evaluate Sentiment with Confidence....29
Apply a Moderation Decision Ladder....29
Manage UGC Rights and Permissions....29
Escalate Reputation Risk with Evidence....30
Build an Advocacy Flywheel....30
Evaluate Organisational Impact....30
Operate an AI Governance and Oversight Loop....31
Lab 9: Triage Community Messages with Human Oversight....31
Lab 10: Capstone: Operate a Governed Agentic Campaign....33
8. Capstone Evidence Checklist....34
9. Sources and Version Notes....35

1. Course Overview

This 2-day, 16-hour WSQ course preserves the approved Social Media Marketing competency coverage while teaching learners to plan, create, measure, optimize and govern social campaigns with bounded AI agents and human oversight.

Learning Outcomes

- LO1: Develop a plan for a Facebook social media marketing campaign.
- LO2: Evaluate social market opportunities for a Facebook campaign.
- LO3: Monitor Facebook social media marketing performance.
- LO4: Determine campaign feasibility by optimizing Facebook advertisements.
- LO5: Evaluate campaign usage and its impact on the organisation.
- LO6: Establish measurable social media marketing objectives.

Assessment

Written Assessment (SAQ): 10 open-ended questions, 60 minutes. Practical Performance: 5 applied tasks, 60 minutes. Both are open book. Submit completed answers on the LMS: https://lms-tms.tertiaryinfotech.com/

2. Responsible Agentic Marketing Operating Model

An agentic workflow observes inputs, plans bounded work, calls approved tools, evaluates outputs, requests human approval for consequential actions and records evidence. Agents support decisions; accountable people retain ownership of objectives, data, claims, budgets, publication and community responses.

Layer	Required evidence	Human control
Objective	KPI tree and guardrails	Campaign owner approves
Data	Purpose, consent, lineage and retention	DPO/policy review
Agent	Prompt, tool contract, model/version and trace	Risk-tier approval
Action	Draft, score, schedule or recommendation	Publish/pause gates
Outcome	Metrics, exceptions and incident record	Review and rollback

3. Topic 01: Plan an Agentic Social Media Campaign

WSQ mapping: A1 · K1 · K2. Legacy coverage retained: Plan a Facebook Social Media Marketing Campaign.

Translate Business Goals into a KPI Tree

Technical anchor: MODEL · K1/A6. The mechanism runs through Business outcome → Marketing objective → Leading signals → Lagging signals → Decision rule.

Field	Worked value	Interpretation
objective_id	OBJ-01	Trace every action

Field	Worked value	Interpretation
baseline	120 leads/month	Starting point
target	+25% in 90 days	Measurable delta
guardrail	CPL <= SGD 24	Economic bound

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Vanity metric	Likes rise but leads do not	Use KPI tree
No baseline	Target cannot be evaluated	Record source period
Agent goal drift	Optimizes clicks only	Lock objective hierarchy
Overclaim	Causal language without test	Label inference

Worked evidence: The 90-day target requires a 25% lift from the 120-lead baseline.

Encode a Persona as Evidence, Not Fiction

Technical anchor: EVIDENCE · K1/A1. The mechanism runs through Collect → Separate → Structure → Score → Approve.

Field	Worked value	Interpretation
persona_id	P-SG-01	Stable key
evidence	12 interviews	Provenance
need	Lower energy bills	Message input
confidence	0.78	Review priority

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Stereotype	Trait lacks evidence	Remove or qualify
PII leakage	Names in prompt	Aggregate and redact
Stale profile	Data older than 12 months	Refresh study
False certainty	No confidence score	Mark hypothesis

Worked evidence: Use multiple evidence sources; no single source is a complete persona.

Model the Buyer Journey as Observable States

Technical anchor: MODEL · K1/A1. The mechanism runs through Unaware → Aware → Considering → Deciding → Customer.

Field	Worked value	Interpretation
journey_state	considering	Current state
trigger	comparison guide	Transition cue
evidence	pricing-page visit	Observable signal
next_best_action	case study	Agent proposal

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Skipped state	Hard sell to unaware user	Use education
Weak signal	One click treated as intent	Require corroboration
Dark pattern	Artificial urgency	Ethics review
State lock	Never re-evaluated	Time-decay signals

Worked evidence: Stage conversion reveals where the agent should intervene, not just final sales.

Build a Social Media Audit Inventory

Technical anchor: ARTIFACT · K1/K2. The mechanism runs through Discover → Verify → Measure → Classify → Assign.

Field	Worked value	Interpretation
asset_id	page_0042	Unique asset
owner	Marketing Ops	Accountability
access	editor	Least privilege
status	fix	Action queue

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Orphaned asset	No owner	Suspend publishing
Excess privilege	Shared admin account	Role-based access
Missing baseline	No comparable window	Export 90 days
Duplicate page	Confused audience	Consolidate

Worked evidence: Most assets can be retained; fixes and retirement need accountable owners.

Score Marketing Opportunities Transparently

Technical anchor: FORMULA · A2/A4. The mechanism runs through Define criteria → Weight → Score → Sensitivity → Recommend.

Field	Worked value	Interpretation
reach	4 × 25%	Audience scale
brand_fit	5 × 30%	Message alignment
cost	3 × 25%	Budget feasibility
risk	4 × 20%	Control burden

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Hidden weighting	Recommendation cannot be audited	Publish rubric
Double count	Reach and impressions overlap	Separate constructs
Agent bias	Favors familiar channels	Blind channel names
Single score	Masks uncertainty	Show sensitivity

Worked evidence: Influencer, paid social, and brand collaboration can be compared with the same rubric.

Design the Marketing Agent Architecture

Technical anchor: ARCH · K3/A1. The mechanism runs through Observe → Plan → Act → Check → Learn.

Field	Worked value	Interpretation
goal	OBJ-01	Mission boundary
tool_allowlist	draft, score	Permitted actions
approval_gate	before_publish	Human control
trace_id	run-2026-0911	Audit evidence

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Over-autonomy	Agent can publish	Remove publish permission

Failure mode	Observable signal	Control / recovery
Tool injection	Content alters instructions	Separate data/instructions
Memory poisoning	Unreviewed feedback persists	Approved writes only
Silent failure	No trace	Structured run log

Worked evidence: Planning, creation, review, and measurement remain explicit stages.

Write a Tool Contract the Agent Can Obey

Technical anchor: CONFIG · K2/K3. The mechanism runs through Name → Input schema → Precondition → Output schema → Error route.

Field	Worked value	Interpretation
tool	draft_post	Clear capability
channel	facebook	Enumerated input
max_claims	3	Bounded output
status	needs_review	Safe default

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Vague tool	Unexpected action	Narrow verb
Free-text input	Unvalidated data	Schema validation
No idempotency	Duplicate posts	request_id
No error class	Infinite retry	Typed errors

Worked evidence: Success, review, validation error, and escalation should be measurable.

Set Policy Boundaries Before Automation

Technical anchor: CONTROL · K2/K4/K5/K6. The mechanism runs through Classify → Permit → Require → Log → Retain.

Field	Worked value	Interpretation
data_class	internal	Handling rule
claim_source	approved catalog	Grounding
reviewer_role	campaign_owner	Accountability
retention_days	180	Lifecycle

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Personal data	Uploaded without consent	Redact/aggregate
IP risk	Unlicensed creative	Rights register
Misleading claim	Unsupported superlative	Evidence requirement
Platform breach	Automation violates terms	Approved interface only

Worked evidence: Most drafts pass; review, revision, and rejection remain visible outcomes.

Lab 1: Build an Agent-Ready Campaign Charter

Mappings: LO1 · LO6 · A1 · A6 · K1 · K2

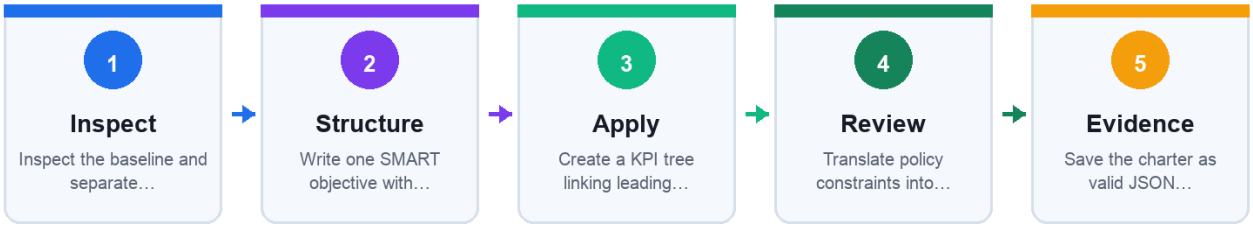
Goal

Use the supplied mock data to produce campaign_charter.json and KPI decision table and demonstrate the mapped abilities under human oversight.

Scenario and deliverable

Lumina Living will launch an energy-saving home kit in Singapore with a 90-day lead target.

Produce: campaign_charter.json and KPI decision table



Before you begin

Open the individual lab folder and inspect every file in mock_data/. Use only the supplied mock data. Create a working copy of each output artifact inside the same lab folder.

Detailed step-by-step procedure

1. Inspect the baseline and separate business outcomes from channel metrics.

Evidence to retain for Step 1: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

2. Write one SMART objective with baseline, target, deadline and owner.

Evidence to retain for Step 2: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

3. Create a KPI tree linking leading indicators to the final outcome.

Evidence to retain for Step 3: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

4. Translate policy constraints into machine-checkable guardrails.

Evidence to retain for Step 4: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

5. Save the charter as valid JSON using the provided schema.

Evidence to retain for Step 5: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

6. Peer-review whether every proposed action is traceable to an objective and policy.

Evidence to retain for Step 6: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

Test it — Verification

- Objective includes baseline 120 leads and a 90-day target.

- CPL guardrail is explicit and numerically testable.
- Publishing permission defaults to human approval.
- No personal data is included in the charter.

Troubleshooting

- If the objective is vague, add a metric, baseline, target and deadline.
- If a guardrail is prose-only, rewrite it as a field/operator/value rule.

Cleanup

- Remove any real client identifiers.
- Keep only the mock files and final charter in the lab folder.

Full lab folder: labs/lab-01-campaign-charter/

Lab 2: Create an Evidence-Based Persona and Journey Agent

Mappings: LO1 · LO2 · A1 · A2 · K1

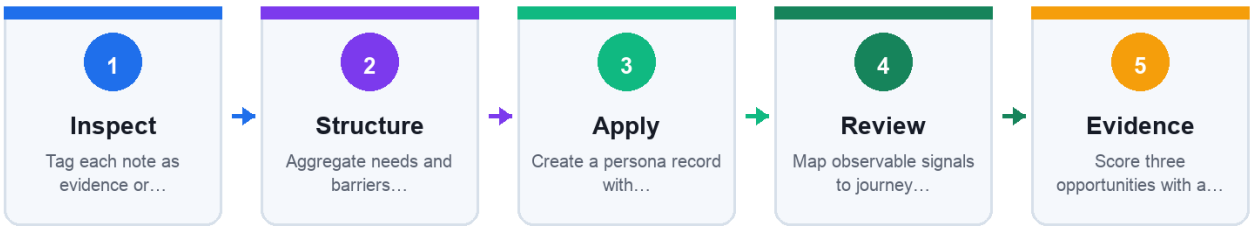
Goal

Use the supplied mock data to produce persona.json, journey.csv and opportunity_score.csv and demonstrate the mapped abilities under human oversight.

Scenario and deliverable

Lumina Living needs a defensible persona and buyer journey for its energy-saving kit.

Produce: persona.json, journey.csv and opportunity_score.csv



Before you begin

Open the individual lab folder and inspect every file in mock_data/. Use only the supplied mock data. Create a working copy of each output artifact inside the same lab folder.

Detailed step-by-step procedure

7. Tag each note as evidence or hypothesis.

Evidence to retain for Step 1: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

8. Aggregate needs and barriers without retaining names.

Evidence to retain for Step 2: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

9. Create a persona record with provenance and confidence fields.

Evidence to retain for Step 3: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

10. Map observable signals to journey states and next-best content.

Evidence to retain for Step 4: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

11. Score three opportunities with a weighted rubric totaling 100%.

Evidence to retain for Step 5: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

12. Document where human validation is required.

Evidence to retain for Step 6: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

Test it — Verification

- Every persona claim points to at least one mock evidence row.
- No invented demographic is presented as fact.
- Journey states use observable signals.
- Opportunity weights total 100%.

Troubleshooting

- If a claim has no source, mark it as a hypothesis.
- If scores tie, run a weight-sensitivity check rather than forcing a winner.

Cleanup

- Delete scratch prompts containing unredacted notes.
- Retain only aggregated persona and journey artifacts.

Full lab folder: labs/lab-02-persona-journey-agent/

4. Topic 02: Create Campaigns with Governed Agents

WSQ mapping: A2 · K3 · K4. Legacy coverage retained: Create a Facebook Social Media Marketing Campaign.

Represent Campaign, Ad Set, and Ad

Technical anchor: ARCH · K8/A1. The mechanism runs through Campaign → Ad set → Ad → Delivery → Result.

Field	Worked value	Interpretation
campaign.objective	LEADS	Optimization goal
adset.budget	SGD 50/day	Spend control
ad.format	vertical_video	Creative contract
event	lead_submitted	Outcome

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Wrong level	Budget set on ad	Respect hierarchy
Objective mismatch	Clicks for lead goal	Optimize for outcome
Missing destination	Broken journey	Preflight URL
Naming drift	No traceability	Naming convention

Worked evidence: Prospecting, retargeting, and experimentation budgets should be explicit.

Turn a Creative Brief into Structured Inputs

Technical anchor: SCHEMA · K1/K3. The mechanism runs through Audience → Promise → Proof → Action → Constraints.

Field	Worked value	Interpretation
audience_id	P-SG-01	Persona link
proof_id	CASE-014	Source trace
cta	Book a demo	Outcome
aspect_ratio	9:16	Placement fit

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Brief stuffing	Too many messages	One promise
Missing proof	Hallucinated claim	Required source ID
Tone conflict	Brand mismatch	Voice rubric
Format mismatch	Crop hides CTA	Safe-zone check

Worked evidence: Message clarity and proof should outrank novelty alone.

Encode Brand Voice as Testable Rules

Technical anchor: CONFIG · K2/K6. The mechanism runs through Collect → Extract → Specify → Test → Approve.

Field	Worked value	Interpretation
warmth	4/5	Desired tone
formality	3/5	Register
banned_phrase	guaranteed	Claims control
reading_level	Grade 8	Accessibility

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Style mimicry	Copies competitor	Use own examples
Ambiguous tone	Reviewer disagreement	Anchored scale
Overconstraint	Robotic copy	Prioritize rules
Unversioned rubric	Inconsistent runs	voice_version

Worked evidence: A measurable rubric separates pass, edit, escalate, and reject outcomes.

Build Audiences with Purpose Limitation

Technical anchor: CONTROL · K5/A2. The mechanism runs through Purpose → Minimize → Transform → Match → Expire.

Field	Worked value	Interpretation
purpose	campaign retargeting	Notification
fields	email_hash	Minimized identifier

Field	Worked value	Interpretation
consent	marketing_opt_in	Legal basis
expiry	2026-12-31	Retention limit

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Purpose creep	List reused	Purpose-bound ID
Raw identifier	Plain email exported	Hash before transfer
No withdrawal	Opt-out ignored	Suppression list
Sensitive inference	Proxy discrimination	Feature review

Worked evidence: Validation and consent filters reduce the usable audience before matching.

Generate Content Through a Review Pipeline

Technical anchor: RUNTIME · K3/K6. The mechanism runs through Retrieve → Draft → Critique → Revise → Approve.

Field	Worked value	Interpretation
brief_id	BR-027	Input trace
variant_count	3	Bounded generation
policy_score	0.94	Automated check
release_status	approved	Human decision

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Ungrounded fact	No source ID	Reject claim
Self-approval	Same agent scores itself	Independent checker
Prompt injection	UGC alters task	Quote as data
Variant collapse	Near duplicates	Similarity threshold

Worked evidence: Only approved variants move from draft to policy pass, human review, and release.

Place a Human Approval Gate Before Publish

Technical anchor: HITL · K2/K4. The mechanism runs through Draft ready → Route → Inspect → Decide → Publish.

Field	Worked value	Interpretation
risk_tier	medium	Routing
review_sla	4 hours	Operational target
decision	approve_with_edit	Outcome
reviewer_id	role:brand_owner	Accountability

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Rubber stamp	Very low review time	Sample audit
No fallback	Reviewer unavailable	Backup role
Post-review mutation	Published copy differs	Content hash
Hidden rejection	No rationale	Reason code

Worked evidence: Edits and rejections are useful control signals, not workflow failures.

Choose Placement and Format with a Matrix

Technical anchor: MODEL · K3/A4. The mechanism runs through Outcome → Audience context → Asset fit → Delivery option → Test.

Field	Worked value	Interpretation
placement	reels	Inventory
ratio	9:16	Creative fit
safe_zone	pass	Message visibility
optimization	lead	Delivery goal

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
One-size asset	CTA cropped	Placement variants
Manual overfit	Delivery constrained	Test automation
Metric mismatch	Views treated as leads	Outcome KPI
No control	Cannot attribute lift	Experiment design

Worked evidence: Short vertical video, carousel, and static image should be evaluated on comparable outcomes.

Design a Clean Campaign Experiment

Technical anchor: FORMULA · K7/A2. The mechanism runs through Hypothesis → Variable → Control → Run → Decide.

Field	Worked value	Interpretation
hypothesis	video improves CVR	Testable claim
primary_metric	conversion_rate	Decision metric
minimum_lift	10%	Practical threshold
stop_rule	>= 500 clicks/cell	Stability

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Peeking	Stop on early winner	Fixed rule
Multiple changes	Causality unclear	One variable
Unequal audience	Biased cells	Random split
Novelty effect	Short run	Adequate duration

Worked evidence: Variant B shows lift, but the sample and decision threshold determine action.

Lab 3: Build a Brand-Safe Content Brief Agent

Mappings: LO2 · A2 · K2 · K3 · K4 · K6

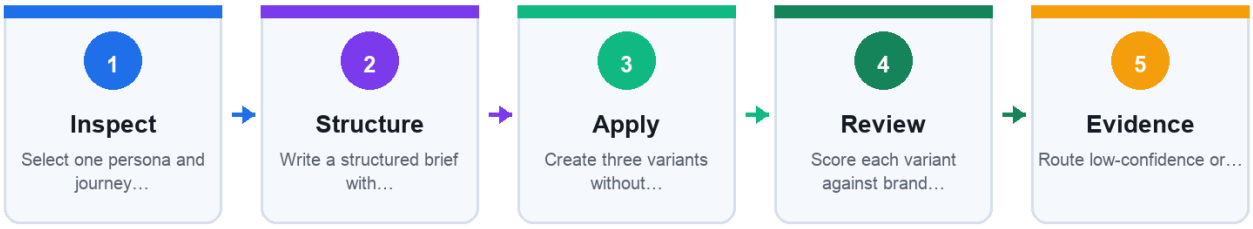
Goal

Use the supplied mock data to produce content_briefs.jsonl and review_rubric.csv and demonstrate the mapped abilities under human oversight.

Scenario and deliverable

Create channel-ready briefs for Lumina Living while preventing unsupported claims and brand drift.

Produce: content_briefs.jsonl and review_rubric.csv



Before you begin

Open the individual lab folder and inspect every file in mock_data/. Use only the supplied mock data. Create a working copy of each output artifact inside the same lab folder.

Detailed step-by-step procedure

13. Select one persona and journey state from Lab 2.

Evidence to retain for Step 1: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

14. Write a structured brief with audience, promise, proof, CTA and constraints.

Evidence to retain for Step 2: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

15. Create three variants without changing the approved fact set.

Evidence to retain for Step 3: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

16. Score each variant against brand voice and policy rubrics.

Evidence to retain for Step 4: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

17. Route low-confidence or unsupported drafts to human review.

Evidence to retain for Step 5: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

18. Record final approval status and content hash.

Evidence to retain for Step 6: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

Test it — Verification

- Each numeric claim carries an approved fact_id.
- Rejected FACT-03 is absent.
- Each brief has one CTA and a defined format.
- Final status is needs_review until a human decision is recorded.

Troubleshooting

- If all variants sound alike, vary angle or format—not factual claims.
- If the rubric is subjective, add anchored examples for scores 1 and 5.

Cleanup

- Remove rejected draft claims from final files.
- Archive the approved version and its source IDs.

Full lab folder: labs/lab-03-content-brief-agent/

Lab 4: Assemble a Governed Campaign Package

Mappings: LO1 · LO2 · A1 · A2 · K3 · K4 · K8

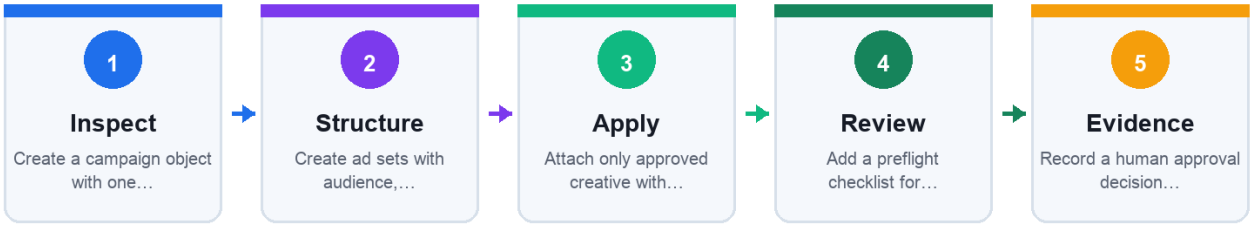
Goal

Use the supplied mock data to produce campaign_package.json and approval_log.csv and demonstrate the mapped abilities under human oversight.

Scenario and deliverable

Translate approved content briefs into a campaign/ad-set/ad package without publishing to a live account.

Produce: campaign_package.json and approval_log.csv



Before you begin

Open the individual lab folder and inspect every file in mock_data/. Use only the supplied mock data. Create a working copy of each output artifact inside the same lab folder.

Detailed step-by-step procedure

19. Create a campaign object with one outcome objective.

Evidence to retain for Step 1: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

20. Create ad sets with audience, budget, schedule and suppression fields.

Evidence to retain for Step 2: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

21. Attach only approved creative with valid usage rights.

Evidence to retain for Step 3: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

22. Add a preflight checklist for destination, tracking and naming.

Evidence to retain for Step 4: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

23. Record a human approval decision with reason and reviewer role.

Evidence to retain for Step 5: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

24. Export a package for simulation only; do not publish.

Evidence to retain for Step 6: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

Test it — Verification

- Campaign, ad set and ad fields are at the correct hierarchy.
- Daily budget stays within SGD 250.
- Pending CR-03 is blocked.
- The package contains no live credential or account ID.

Troubleshooting

- If delivery settings conflict, trace which hierarchy owns the field.
- If a creative is blocked, check status and rights_expiry.

Cleanup

- Delete any accidental real account identifiers.
- Keep the approved simulation package and decision log.

Full lab folder: labs/lab-04-campaign-builder/

5. Topic 03: Measure Performance and Agent Decisions

WSQ mapping: A3 · K5 · K6. Legacy coverage retained: Measure Facebook Social Media Marketing Performance.

Map the Measurement Event Pipeline

Technical anchor: ARCH · A3. The mechanism runs through Impression → Click → Session → Conversion → Reconcile.

Field	Worked value	Interpretation
event_name	lead_submitted	Stable taxonomy
event_time	ISO-8601	Sequence
campaign_id	CMP-26-04	Join key
value	120.00 SGD	Outcome value

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Duplicate event	Inflated conversions	event_id dedupe
Missing consent	Illegal collection	Consent gate
Clock drift	Wrong attribution	UTC timestamps
Join failure	No campaign ID	Required key

Worked evidence: Loss between stages is expected; unexplained discontinuities are instrumentation defects.

Calculate Reach, CTR, CVR, CPL and ROAS

Technical anchor: FORMULA · A3. The mechanism runs through Reach → CTR → CVR → CPL → ROAS.

Field	Worked value	Interpretation
impressions	50,000	CTR denominator
clicks	3,400	CTR = 6.8%
leads	96	CVR = 2.82%
spend	SGD 2,400	CPL = SGD 25

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Mixed windows	Spend and leads differ	Same date range
Wrong denominator	CVR uses impressions	Metric dictionary
Gross revenue	ROAS overstates margin	Add contribution margin
Tiny sample	Volatile rate	Show counts

Worked evidence: Rates are meaningful only when their underlying counts and windows are visible.

Build a Funnel with Diagnostic Rates

Technical anchor: MODEL · A3. The mechanism runs through Exposure → Interest → Intent → Lead → Value.

Field	Worked value	Interpretation
stage	lead	Named state
count	96	Observed total
rate_from_prior	23.4%	Local conversion
drop_reason	form friction	Diagnosis

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Double counting	Repeated events	Unique key
Unknown loss	No reason codes	Instrument exits
Quality ignored	Cheap low-value leads	Qualification rate
Lag ignored	Recent leads not matured	Cohort window

Worked evidence: A low early-stage rate and a strong late-stage rate imply a targeting or creative issue.

Separate Attribution from Incrementality

Technical anchor: CASE · A3. The mechanism runs through Observed conversion → Attribution rule → Counterfactual → Experiment → Decision.

Field	Worked value	Interpretation
window	7-day click	Attribution scope
credited	120	Reported conversions
holdout	24	Baseline conversions
incremental	96	Estimated lift

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Last-click bias	Upper funnel erased	Compare models
View-through inflation	Passive exposure credited	Separate view/click
No counterfactual	Correlation claimed causal	Holdout
Cross-device gap	Identity incomplete	State limitation

Worked evidence: Attributed conversions can exceed estimated incremental conversions.

Design a Reporting Schema for Agents

Technical anchor: SCHEMA · K8/A3. The mechanism runs through Extract → Normalize → Compute → Explain → Deliver.

Field	Worked value	Interpretation
report_date	2026-09-11	As-of time
currency	SGD	Comparable spend
formula_version	metrics-v2	Reproducibility
source_rows	campaign.csv:48	Lineage

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Silent refresh	Numbers change	Snapshot timestamp
Schema drift	Column renamed	Validation
Rounding error	Totals disagree	Compute then format
Narrative hallucination	Claim lacks row	Evidence citation

Worked evidence: Automation can reduce preparation time while review and sign-off remain explicit.

Detect Anomalies with Thresholds and Context

Technical anchor: FORMULA · A3. The mechanism runs through Baseline → Band → Trigger → Context → Escalate.

Field	Worked value	Interpretation
metric	CPL	Signal
baseline	SGD 22	Reference
threshold	> SGD 30	Trigger
severity	high	Routing

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Seasonality	Normal spike flagged	Comparable baseline
Alert flood	Every metric triggers	Priority tiers
Auto-pause harm	False positive stops winner	Human approval
No context	Budget change ignored	Event annotations

Worked evidence: The final-day spike breaches the SGD 30 decision threshold and needs diagnosis.

Build Retargeting with Consent and Suppression

Technical anchor: CONTROL · K5/A3. The mechanism runs through Qualify → Segment → Suppress → Sequence → Expire.

Field	Worked value	Interpretation
segment	cart_7d	Behavior cohort
consent	true	Eligibility
suppression	purchased_30d	Exclusion
ttl_days	7	Expiry

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
No suppression	Customers see acquisition ad	Exclude converters
Long retention	Stale tracking	TTL
Sensitive proxy	Inferred condition	Prohibited features
Frequency harm	Excess exposure	Cap and monitor

Worked evidence: Consent, suppression, and freshness filters reduce the addressable group.

Observe the Agent, Not Just the Campaign

Technical anchor: EVIDENCE · A3. The mechanism runs through Trace → Record → Score → Review → Improve.

Field	Worked value	Interpretation
trace_id	run-0911-042	Correlation
tool_calls	6	Action count
policy_flags	1	Risk signal
final_status	approved	Outcome

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
No trace	Cannot reproduce	Mandatory run ID
Sensitive logs	PII retained	Redaction
Success-only metrics	Failures hidden	All outcomes
Unsafe learning	Bad output saved	Reviewed memory

Worked evidence: Approval, revision, validation failure, and escalation should be tracked separately.

Lab 5: Compute a Campaign Measurement Scorecard

Mappings: LO3 · A3 · K5 · K8

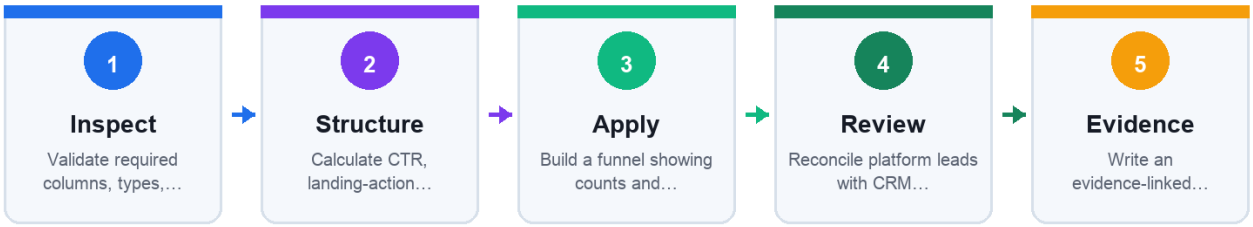
Goal

Use the supplied mock data to produce scorecard.csv, metric_dictionary.md and funnel_analysis.md and demonstrate the mapped abilities under human oversight.

Scenario and deliverable

Reconcile a mock platform export and CRM outcome table, then calculate decision-ready metrics.

Produce: scorecard.csv, metric_dictionary.md and funnel_analysis.md



Before you begin

Open the individual lab folder and inspect every file in mock_data/. Use only the supplied mock data. Create a working copy of each output artifact inside the same lab folder.

Detailed step-by-step procedure

25. Validate required columns, types, currency and date window.

Evidence to retain for Step 1: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

26. Calculate CTR, landing-action rate, CVR, CPL and ROAS.

Evidence to retain for Step 2: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

27. Build a funnel showing counts and stage-to-stage rates.

Evidence to retain for Step 3: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

28. Reconcile platform leads with CRM quality outcomes.

Evidence to retain for Step 4: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

29. Write an evidence-linked interpretation with limitations.

Evidence to retain for Step 5: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

30. Define scale, hold and stop decision rules.

Evidence to retain for Step 6: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

Test it — Verification

- CTR equals clicks divided by impressions.
- CPL equals spend divided by leads.
- ROAS is clearly labeled as revenue/ad spend.
- Every narrative claim cites a computed row.

Troubleshooting

- If totals disagree, confirm the same campaign and date window.
- If a rate is unstable, show its raw count and avoid overclaiming.

Cleanup

- Remove any formula scratch sheets with broken references.
- Retain the clean input, metric dictionary and scorecard.

Full lab folder: labs/lab-05-measurement-scorecard/

Lab 6: Build a Listening and Anomaly Triage Agent

Mappings: LO3 · A3 · K8 · K10

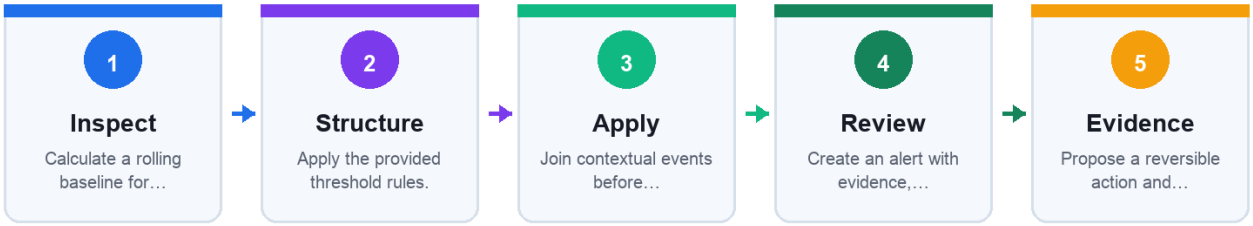
Goal

Use the supplied mock data to produce alerts.csv and incident_triage.md and demonstrate the mapped abilities under human oversight.

Scenario and deliverable

Detect a CPL spike and rising negative feedback without allowing the agent to pause live campaigns automatically.

Produce: alerts.csv and incident_triage.md



Before you begin

Open the individual lab folder and inspect every file in mock_data/. Use only the supplied mock data. Create a working copy of each output artifact inside the same lab folder.

Detailed step-by-step procedure

31. Calculate a rolling baseline for each signal.

Evidence to retain for Step 1: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

32. Apply the provided threshold rules.

Evidence to retain for Step 2: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

33. Join contextual events before assigning likely cause.

Evidence to retain for Step 3: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

34. Create an alert with evidence, severity and owner.

Evidence to retain for Step 4: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

35. Propose a reversible action and rollback condition.

Evidence to retain for Step 5: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

36. Record that a human—not the agent—decides whether to pause.

Evidence to retain for Step 6: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

Test it — Verification

- The 2026-09-05 CPL and comment alerts fire.
- The budget increase is retained as context.
- No rule authorizes automatic campaign pause.
- The alert includes source rows and owner.

Troubleshooting

- If every day alerts, compare against a stable baseline and tune severity.
- If no alert fires, check numeric types and operators.

Cleanup

- Delete test alerts marked as real incidents.
- Retain the rules, mock data and final triage record.

Full lab folder: labs/lab-06-listening-anomaly-agent/

6. Topic 04: Optimize Campaigns with Controlled Experiments

WSQ mapping: A4 · K7 · K8. Legacy coverage retained: Optimize Facebook Social Media Marketing Performance.

Run the Observe–Diagnose–Act Loop

Technical anchor: RUNTIME · K7/A4. The mechanism runs through Observe → Diagnose → Propose → Approve → Measure.

Field	Worked value	Interpretation
signal	CTR -22%	Observed change
hypothesis	creative fatigue	Diagnosis
action	rotate variant B	Intervention
rollback	CTR < baseline	Safety

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Metric chasing	Changes daily	Minimum window
Wrong diagnosis	Budget cut for tracking bug	Data health first
Large action	100% reallocation	Bounded step
No rollback	Loss compounds	Precommit rule

Worked evidence: Automation shortens observation and diagnosis but approval remains deliberate.

Use Practical Significance in A/B Decisions

Technical anchor: FORMULA · K7/A4. The mechanism runs through Set baseline → Set minimum lift → Collect sample → Compare → Decide.

Field	Worked value	Interpretation
baseline_cvr	4.8%	Control
variant_cvr	5.3%	Observed
relative_lift	10.4%	Effect
min_lift	10%	Business threshold

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Stat-only win	Tiny useless lift	Practical threshold
Underpowered	Wide uncertainty	Extend sample
Metric switch	Choose winner after seeing data	Pre-register
Segment mining	False positives	Limit comparisons

Worked evidence: The observed lift barely clears the business threshold; uncertainty still matters.

Reallocate Budget with Marginal Return

Technical anchor: FORMULA · A4. The mechanism runs through Measure → Estimate → Rank → Move → Recheck.

Field	Worked value	Interpretation
channel	retargeting	Budget pool
marginal_cpl	SGD 18	Next-dollar cost
max_shift	15%	Change bound
review_after	48 hours	Stability

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Average return	Hides saturation	Use marginal
Large shift	Resets learning	Cap change
Delayed value	Leads not matured	Cohort lag
Automated overspend	No hard ceiling	Account budget cap

Worked evidence: Budget should move toward the lowest marginal CPL within delivery and risk constraints.

Detect Creative Fatigue

Technical anchor: EVIDENCE · K7/A4. The mechanism runs through Track → Compare → Confirm → Rotate → Learn.

Field	Worked value	Interpretation
frequency	4.6	Exposure load
ctr_change	-28%	Response decay
comment_tone	negative rising	Qualitative signal
action	rotate	Decision

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
False fatigue	Tracking outage	Data health check
Audience depletion	Small pool	Broaden carefully
Unapproved asset	Fast rotation skips review	Preapproved library
Learning loss	Old asset deleted	Archive version

Worked evidence: Declining CTR as frequency rises is a fatigue signal, not proof by itself.

Control Audience Overlap and Frequency

Technical anchor: MODEL · K7/A4. The mechanism runs through Map → Estimate overlap → Prioritize → Exclude → Monitor.

Field	Worked value	Interpretation
audience_a	prospects_sg	Primary
audience_b	site_30d	Retarget
overlap	38%	Conflict signal
frequency_cap	3/7d	Exposure guard

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Self-competition	CPM rises	Mutual exclusions
Overexposure	Negative feedback	Frequency guard
Too narrow	Delivery stalls	Minimum size
Proxy targeting	Fairness risk	Feature review

Worked evidence: The 38% pair needs exclusions or a clear prioritization rule.

Schedule Content with Dependencies

Technical anchor: CONFIG · K8/A4. The mechanism runs through Backlog → Calendar → Dependency → Publish → Reconcile.

Field	Worked value	Interpretation
content_id	CNT-118	Stable item
publish_at	2026-09-15T10:00+08	Timezone
dependency	legal-approved	Gate
request_id	pub-118-v2	No duplicates

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Timezone error	Publishes eight hours late	Explicit offset
Duplicate post	Retry creates copy	Idempotency
Expired creative	Promotion ended	Validity window
Missing approval	Queue bypass	Hard dependency

Worked evidence: Education, proof, community, and promotion should reflect the campaign objective.

Compare Platforms with a Feasibility Matrix

Technical anchor: MODEL · K3/A4. The mechanism runs through Need → Criteria → Evidence → Score → Pilot.

Field	Worked value	Interpretation
platform	Facebook	Option
audience_fit	4/5	Evidence score
measurement	4/5	Data capability
pilot_budget	SGD 500	Reversible test

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Popularity bias	Largest platform wins	Outcome weighting
Stale demographics	Old report used	As-of date
Apples/oranges	Different KPIs	Common outcome
Full rollout	No pilot	Test first

Worked evidence: Platform choice combines fit, measurement, format, cost, and governance—not reach alone.

Define Rollback and Kill-Switch Rules

Technical anchor: CONTROL · K2/A4. The mechanism runs through Bound → Trigger → Pause → Restore → Review.

Field	Worked value	Interpretation
max_daily_spend	SGD 250	Hard ceiling
kill_trigger	policy_severity=critical	Immediate stop
rollback_version	campaign-v8	Known good
owner	marketing_director	Authority

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Soft limit	Agent overrides	Enforce externally
No snapshot	Cannot restore	Version configuration
Broad kill	All campaigns paused	Scope by campaign
Restart loop	Failure repeats	Human reopen

Worked evidence: Spend, CPL, budget-shift, and critical-policy thresholds constrain autonomy.

Lab 7: Evaluate an A/B Test and Budget Decision

Mappings: LO4 · A4 · K7

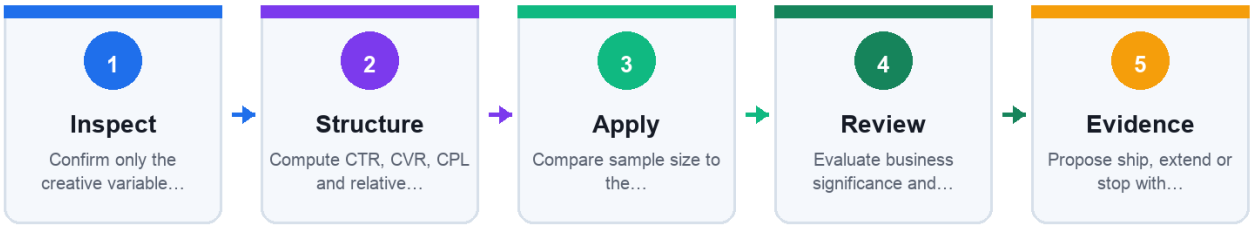
Goal

Use the supplied mock data to produce experiment_decision.md and budget_change.json and demonstrate the mapped abilities under human oversight.

Scenario and deliverable

Determine whether creative B should receive more budget without reacting to an early, unstable result.

Produce: experiment_decision.md and budget_change.json



Before you begin

Open the individual lab folder and inspect every file in mock_data/. Use only the supplied mock data. Create a working copy of each output artifact inside the same lab folder.

Detailed step-by-step procedure

37. Confirm only the creative variable differs.

Evidence to retain for Step 1: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

38. Compute CTR, CVR, CPL and relative lift for both cells.

Evidence to retain for Step 2: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

39. Compare sample size to the precommitted rule.

Evidence to retain for Step 3: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

40. Evaluate business significance and uncertainty.

Evidence to retain for Step 4: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

41. Propose ship, extend or stop with rationale.

Evidence to retain for Step 5: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

42. If shipping, create a bounded budget change and rollback rule.

Evidence to retain for Step 6: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

Test it — Verification

- Both cells exceed 500 clicks.
- Relative CVR lift is calculated from the control baseline.
- Budget change does not exceed 15%.
- The decision states evidence and uncertainty.

Troubleshooting

- If the lift looks huge, verify percentages use the correct denominator.
- If the test changed multiple variables, mark the causal conclusion invalid.

Cleanup

- Remove exploratory metrics chosen after the result.
- Retain the precommitted policy and final decision.

Full lab folder: labs/lab-07-ab-test-optimizer/

Lab 8: Orchestrate an Approval-Aware Content Calendar

Mappings: LO4 · A4 · K4 · K8

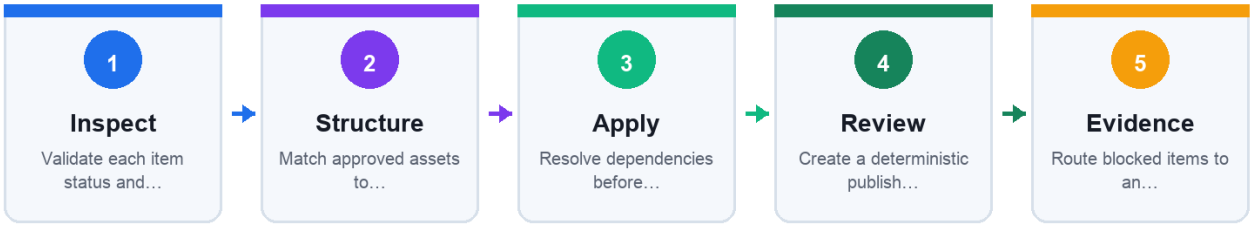
Goal

Use the supplied mock data to produce approved_calendar.csv and exception_queue.csv and demonstrate the mapped abilities under human oversight.

Scenario and deliverable

Schedule an eight-item campaign calendar with timezones, dependencies, rights expiry and idempotent publishing requests.

Produce: approved_calendar.csv and exception_queue.csv



Before you begin

Open the individual lab folder and inspect every file in mock_data/. Use only the supplied mock data. Create a working copy of each output artifact inside the same lab folder.

Detailed step-by-step procedure

43. Validate each item status and validity window.

Evidence to retain for Step 1: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

44. Match approved assets to compatible slots and formats.

Evidence to retain for Step 2: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

45. Resolve dependencies before scheduling.

Evidence to retain for Step 3: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

46. Create a deterministic publish request_id.

Evidence to retain for Step 4: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

47. Route blocked items to an exception queue.

Evidence to retain for Step 5: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

48. Reconcile the planned schedule with mock actual post IDs.

Evidence to retain for Step 6: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

Test it — Verification

- Pending CNT-103 is not scheduled.
- All timestamps include +08:00.
- No content is scheduled after valid_to.
- Every scheduled row has a unique request_id.

Troubleshooting

- If a duplicate appears, check request_id construction.
- If a slot is empty, inspect approval, rights and format constraints.

Cleanup

- Remove any live webhook or platform credentials.
- Retain the final calendar and exception queue.

Full lab folder: labs/lab-08-content-calendar-orchestrator/

7. Topic 05: Manage Communities with Human Oversight

WSQ mapping: A5 · A6 · K9 · K10. Legacy coverage retained: Manage Facebook Social Media Community.

Route Community Messages by Intent and Risk

Technical anchor: RUNTIME · K9/K10. The mechanism runs through Ingest → Classify → Score → Route → Close.

Field	Worked value	Interpretation
intent	delivery_issue	Work queue
sentiment	negative	Context
risk_score	0.82	Escalation
sla_minutes	30	Response target

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Sarcasm	Sentiment misread	Human review
Private data	Order number public	Move to private channel
Wrong queue	Crisis treated routine	Severity rules
No closure	Repeated complaint	Case status

Worked evidence: Routine, assisted, escalated, and rejected items need distinct workflows.

Evaluate Sentiment with Confidence

Technical anchor: MODEL · K10/A5. The mechanism runs through Normalize → Classify → Estimate → Sample → Adjust.

Field	Worked value	Interpretation
label	negative	Prediction
confidence	0.64	Uncertainty
language	en-SG	Context
review	required	Low-confidence route

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Dialect error	Singlish misread	Local sample
Aggregate masking	Crisis subgroup hidden	Segment view
Confidence misuse	0.51 auto-action	Threshold
Feedback leakage	Reviewer label includes PII	Redact

Worked evidence: Distribution is a signal; confidence and representative sampling determine trust.

Apply a Moderation Decision Ladder

Technical anchor: CONTROL · K9/K10. The mechanism runs through Allow → Assist → Review → Escalate → Remove.

Field	Worked value	Interpretation
policy_code	HARASS-02	Rule
severity	high	Priority
evidence_hash	sha256:…	Integrity
decision	escalate	Outcome

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Over-removal	Legitimate criticism hidden	Appeal route
Under-removal	Harm remains	High-risk threshold
Context loss	Quoted slur punished	Conversation window
No evidence	Decision disputed	Capture snapshot

Worked evidence: Most content remains visible; sensitive decisions need evidence and appeal.

Manage UGC Rights and Permissions

Technical anchor: ARTIFACT · K6/K10. The mechanism runs through Discover → Request → Record → Credit → Expire.

Field	Worked value	Interpretation
ugc_id	UGC-204	Asset key
creator	@sample_creator	Rights holder

Field	Worked value	Interpretation
channels	Facebook, Instagram	Licensed scope
expires	2026-12-31	Term

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Hashtag assumption	Tag treated as permission	Ask explicitly
Scope creep	Use in paid ad	Record channels
Missing credit	Creator omitted	Attribution field
Expired right	Asset still scheduled	Calendar validation

Worked evidence: Approved, pending, declined, and expired assets must remain distinguishable.

Escalate Reputation Risk with Evidence

Technical anchor: CASE · K9/A5. The mechanism runs through Detect → Verify → Contain → Respond → Review.

Field	Worked value	Interpretation
mentions_15m	84	Velocity
severity	critical	Priority
owner	communications_lead	Authority
next_update	30 min	Cadence

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Auto-defence	Agent argues publicly	Pause replies
Slow escalation	Issue compounds	Velocity trigger
Unverified claim	Rumor repeated	Source check
No timeline	Conflicting updates	Incident log

Worked evidence: Acceleration, not just volume, can trigger a crisis escalation.

Build an Advocacy Flywheel

Technical anchor: MODEL · K9/A5. The mechanism runs through Listen → Recognize → Enable → Amplify → Measure.

Field	Worked value	Interpretation
advocate_id	ADV-031	Relationship key
contribution	tutorial video	Value
permission	organic repost	Scope
referrals	7	Outcome

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Transactional reward	Community feels bought	Value exchange
Favoritism	Same voices amplified	Diversity review
Consent gap	Repost without permission	Rights record
Vanity advocacy	Reach only	Retention/referral

Worked evidence: Recognition can progress from contributors to repeat advocates and referrals.

Evaluate Organisational Impact

Technical anchor: FORMULA · A5/A6. The mechanism runs through Define → Baseline → Measure → Attribute → Decide.

Field	Worked value	Interpretation
revenue_lift	SGD 18,000	Outcome
cost	SGD 7,200	Investment
complaint_rate	1.8%	Guardrail
confidence	medium	Evidence strength

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Single metric	Sales rise, trust falls	Balanced scorecard
No counterfactual	All lift credited	State limits
Hidden labor	Review time ignored	Full cost
Short window	No retention view	Cohort follow-up

Worked evidence: Financial value, cost, satisfaction lift, and complaint rate tell different parts of impact.

Operate an AI Governance and Oversight Loop

Technical anchor: GOVERNANCE · K2/K5/A5. The mechanism runs through Register → Assess → Control → Monitor → Retire.

Field	Worked value	Interpretation
use_case_id	AI-SMM-01	Inventory
risk_tier	medium	Control depth
human_owner	marketing_director	Accountability
review_cycle	quarterly	Oversight

Failure, control and recovery

Failure mode	Observable signal	Control / recovery
Shadow agent	No registered owner	Inventory
Data drift	Audience changes	Monitoring
Vendor change	Behavior shifts	Change review
Zombie access	Retired tool active	Revoke credentials

Worked evidence: Registration, risk assessment, control testing, and review completion form an evidence chain.

Lab 9: Triage Community Messages with Human Oversight

Mappings: LO5 · A5 · K5 · K9 · K10

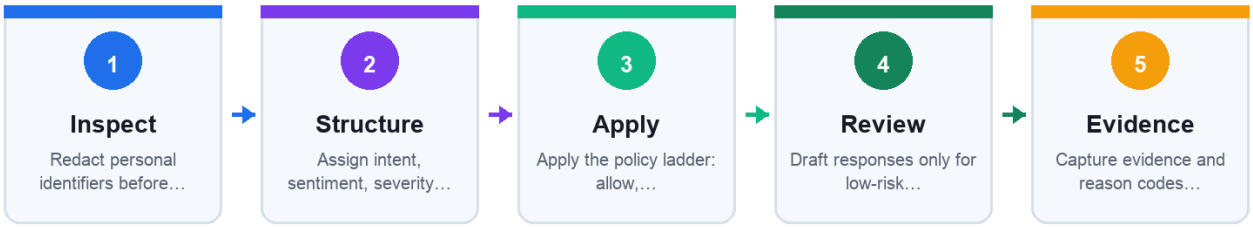
Goal

Use the supplied mock data to produce moderation_queue.csv and response_log.md and demonstrate the mapped abilities under human oversight.

Scenario and deliverable

Classify and route a queue of comments while protecting personal data and preserving legitimate criticism.

Produce: moderation_queue.csv and response_log.md



Before you begin

Open the individual lab folder and inspect every file in mock_data/. Use only the supplied mock data. Create a working copy of each output artifact inside the same lab folder.

Detailed step-by-step procedure

49. Redact personal identifiers before AI-assisted classification.

Evidence to retain for Step 1: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

50. Assign intent, sentiment, severity and confidence.

Evidence to retain for Step 2: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

51. Apply the policy ladder: allow, assist, review, escalate or remove.

Evidence to retain for Step 3: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

52. Draft responses only for low-risk items.

Evidence to retain for Step 4: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

53. Capture evidence and reason codes for sensitive decisions.

Evidence to retain for Step 5: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

54. Review a sample for false positives and local-language errors.

Evidence to retain for Step 6: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

Test it — Verification

- Order number in M002 is redacted before processing.
- M004 is emergency-escalated and not auto-replied.
- M003 remains visible while routed for human review.
- Every decision has a reason code.

Troubleshooting

- If sarcasm or dialect is uncertain, lower confidence and route to review.
- If criticism is removed merely for being negative, revisit the policy criterion.

Cleanup

- Delete unredacted scratch copies.
- Retain only the redacted queue and evidence log.

Full lab folder: labs/lab-09-community-moderation-agent/

Lab 10: Capstone: Operate a Governed Agentic Campaign

Mappings: LO1–LO6 · A1–A6 · K1–K10

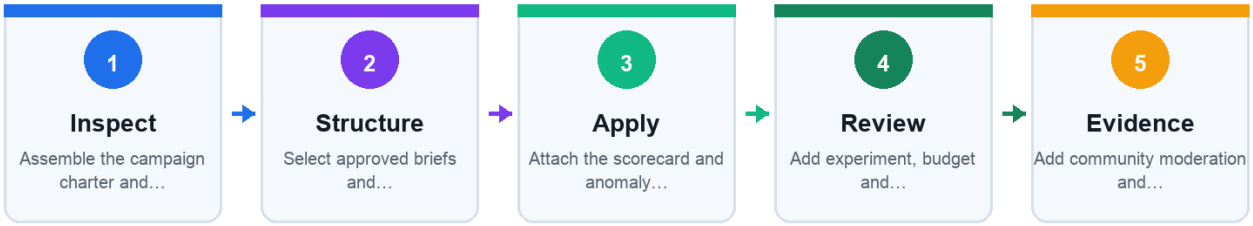
Goal

Use the supplied mock data to produce campaign_runbook.md, governance_register.csv and executive_review.md and demonstrate the mapped abilities under human oversight.

Scenario and deliverable

Integrate planning, creation, measurement, optimization and community controls into one auditable Lumina Living campaign runbook.

Produce: campaign_runbook.md, governance_register.csv and executive_review.md



Before you begin

Open the individual lab folder and inspect every file in mock_data/. Use only the supplied mock data. Create a working copy of each output artifact inside the same lab folder.

Detailed step-by-step procedure

55. Assemble the campaign charter and persona evidence from Labs 1–2.

Evidence to retain for Step 1: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

56. Select approved briefs and campaign configuration from Labs 3–4.

Evidence to retain for Step 2: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

57. Attach the scorecard and anomaly rules from Labs 5–6.

Evidence to retain for Step 3: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

58. Add experiment, budget and calendar controls from Labs 7–8.

Evidence to retain for Step 4: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

59. Add community moderation and crisis escalation from Lab 9.

Evidence to retain for Step 5: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

60. Run a tabletop simulation and record human decisions, failures and rollback actions.

Evidence to retain for Step 6: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

61. Write an executive review covering value, risk, evidence and next action.

Evidence to retain for Step 7: the changed field, formula, decision rationale or approval record. Do not use a screenshot as a substitute for editable evidence.

Test it — Verification

- Every agent action has an owner, input, output and control.
- All six abilities and ten knowledge criteria are evidenced.
- No live credentials or personal data exist in the package.
- The runbook contains a kill switch, rollback version and incident route.

Troubleshooting

- If an artifact cannot be traced, add its source lab and version.
- If the agent can publish or pause without approval, narrow its tool contract.

Cleanup

- Remove all real names, credentials and platform account IDs.
- Archive one final immutable capstone package for assessment evidence.

Full lab folder: labs/lab-10-capstone-agentic-campaign/

8. Capstone Evidence Checklist

- Objectives and KPI rules are measurable.
- Persona and audience claims have provenance and confidence.
- Campaign objects respect hierarchy and policy.
- Every generated claim cites approved evidence.
- Metrics use consistent windows and denominators.
- Optimization changes are bounded and reversible.
- Community decisions preserve privacy, rights and appeal.
- Human decisions and agent actions are traceable.

9. Sources and Version Notes

- Legacy coverage: WSQ Master Trainer Slides v14 supplied in reference/
- QuickBook: Agent-Powered Growth, John Wiley & Sons, 2026 — architecture, deployment and scaling concepts paraphrased
- QuickBook: AI in Marketing: Applications, Insights, and Analysis, Routledge, 2025 — applications, consumer insight and ethics concepts paraphrased
- Course page: https://www.tertiarycourses.com.sg/wsq-agentic-ai-for-social-media-marketing.html
- Meta for Business: https://www.facebook.com/business/ads/facebook-instagram-reels-ads
- PDPC obligations: https://www.pdpc.gov.sg/overview-of-pdpa/the-legislation/personal-data-protection-act/data-protection-obligations